Introduction

STAT 3160: Applied Multivariate Analysis

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Objectives

- Multivariate Data: When studying complex phenomenon, we need to collect measurements (observations) on many different variables.
- Multivariate techniques: methods to elicit information from multivariate data. Most are statistical.
- Sorting and grouping: create groups of "similar" objects or classify objects into well-defined groups
- Investigation of the dependence among variables.
- Prediction.
- Hypothesis testing $\rightarrow$ validate substantive theory.

Population Parameters and Sample Statistics

Statistics, as a subject matter, is the science and art of using sample information to make generalizations about populations.

- A **population** is the collection of all people, plants, animals, or objects of interest about which we wish to make statistical inferences (generalizations).
- A **population parameter** is a numerical characteristic of a population. In nearly all statistical problems we do not know the value of a parameter because we do not measure the entire population. We use sample data to make an inference about the value of a parameter.
- A **sample** is the subset of the population that we actually measure or observe.
- A **sample statistic** is a numerical characteristic of a sample. A sample statistic estimates the unknown value of a population parameter. Information collected from sample statistic is sometimes refereed to as Descriptive Statistic.

Introduction

A partial description of the joint distribution of the data is provided here. Three aspects of the data are of importance, the first two of which you should already be familiar with from univariate statistics. These are:

- Central Tendency. What is a typical value for each variable?
- Dispersion. How far apart are the individual observations from a central value for a given variable?
- Association. This might (or might not!) be a new measure for you. When more than one variable are studies together how does each variable relate to the remaining variables? How are the variables simultaneously related to one another? Are they positively or negatively related?

Statistics Review

Getting use to the notation we'll use.

- Data: measurements on p variables (attributes, characteristics, etc) for each of n cases (individuals, experimental units, etc).
- Notation:
  1. x_{jk} = measurement of the kth variable on the jth case.
  2. $j = 1, \ldots, n$ where n = the number of cases.
  3. $k = 1, \ldots, p$ where p = the number of variables.

Data Array

Data are arranged as an array or matrix of cases by variables

case	Variable 1	Variable 2	...	Variable k	...	Variable p
1	x_{11}	x_{12}	...	x_{1k}	...	x_{1p}
2	x_{21}	x_{22}	...	x_{2k}	...	x_{2p}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
j	x_{j1}	x_{j2}	...	x_{jk}	...	x_{jp}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	x_{n1}	x_{n2}	...	x_{nk}	...	x_{np}

This is a $n \times p$ matrix X.

Measures of Central Tendency

In multivariate statistics we will always be working with vectors of observations. So in this case we are going to arrange the data for the p variables on each subject into a vector. In the expression below, $\mathbf{X}$ is the vector of p random variables. Therefore, the data for the jth variable will be located in the jth element of this subject's vector, $j = 1$ to p

$$\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix}$$

The population mean is the measure of central tendency for the population. Here, the population mean for variable j is

$$\mu_j = E(X_j).$$

Measures of Dispersion

A variance measures the degree of spread (dispersion) in a variable's values. Theoretically, a population variance is the average squared difference between a variable's values and the mean for that variable.

The population variance for variable x_j is

$$\sigma_j^2 = E(x_j - \mu_j)^2.$$

Measures of Association

Association is concerned with how each variable is related to the other variable(s). In this case the first measure that we will consider is the covariance between two variables j and k.
The **population covariance**: the population covariance between variables j and k is

$$\sigma_{jk} = E(X_j - \mu_j)(X_k - \mu_k)$$

Measures of Dispersion

The magnitude of the covariance value is not particularly helpful as it is a function of the magnitudes (scales) of the two variables.
Correlation: This suggests an alternative measure of association. The population correlation is defined to be equal to the population covariance divided by the product of the population standard deviations:

$$\rho_{jk} = \frac{\sigma_{jk}}{\sigma_j \sigma_k}$$

Sample Estiamtion

Suppose we have n observations on one variable:

$$x_{11}, x_{21}, \ldots, x_{n1} \quad \text{(1st column of } X\text{)}$$

- Mean

$$\bar{x}_1 = \frac{1}{n}\sum_{j=1}^{n} x_{j1}$$

- If the n observations are a sample from a larger population of possible measurements, then $\bar{x}_1$ is the *sample mean*.
- In general, sample means

$$\bar{x}_k = \frac{1}{n}\sum_{j=1}^{n} x_{jk}, \quad \text{for} \quad k = 1, \ldots, p$$

This is the mean of n observations in the kth column of X.

Variance & Standard Deviation

A sample of n observations on one variable from some population:

$$x_{11}, x_{21}, \ldots, x_{n1}$$

- **The sample variance** of the first variable is

$$s_1^2 = \frac{1}{n-1} \sum_{j=1}^{n} (x_{j1} - \bar{x}_1)^2 = s_{11}$$

- In general,

$$s_{kk} \equiv s_k^2 = \frac{1}{n-1} \sum_{j=1}^{n} (x_{jk} - \bar{x}_k)^2,$$

  where $k = 1, \ldots, p$.

- **Sample standard deviation** is $\sqrt{s_{kk}} = s_k$.

Covaraince

- A sample of n observations on two variables from some population

$$\underbrace{\begin{pmatrix} x_{11} \\ x_{12} \end{pmatrix}}_{1^{st}\,case}, \underbrace{\begin{pmatrix} x_{21} \\ x_{22} \end{pmatrix}}_{2^{nd}\,case}, \ldots, \underbrace{\begin{pmatrix} x_{n1} \\ x_{n2} \end{pmatrix}}_{n^{th}\,case}$$

- **Sample covariance**

$$s_{12} = \frac{1}{n-1} \sum_{j=1}^{n} (x_{j1} - \bar{x}_1)(x_{j2} - \bar{x}_2)$$

- Properties:
  1. $s_{12} > 0 \rightarrow$ large values tend to occur together
  2. $s_{12} < 0 \rightarrow$ larger values tend to occur with small values
  3. $s_{12} = 0 \rightarrow$ variables are not linearly associated

Covariance(continued)

In general,

$$s_{ik} = \frac{1}{n-1} \sum_{j=1}^{n} (x_{ji} - \bar{x}_i)(x_{jk} - \bar{x}_k)$$

- When $i = k$, $s_{ii} = s_i^2 = i$th sample variance.
- Covariances are symmetric: $s_{ik} = s_{ki}$.
- An array of variances and covariances:

Variable	1	2	...	p
1	s_{11}	s_{12}	...	s_{1p}
2	s_{12}	s_{22}	...	s_{2p}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
p	s_{1p}	s_{2p}	...	s_{pp}

Sample Correlation Coefficients

$$r_{ik} = \frac{s_{ik}}{\sqrt{s_{ii}}\sqrt{s_{kk}}} = \frac{\sum_{j=1}^{n}(x_{ji} - \bar{x}_i)(x_{jk} - \bar{x}_k)}{\sqrt{\sum_{j=1}^{n}(x_{ji} - \bar{x}_i)^2}\sqrt{\sum_{j=1}^{n}(x_{jk} - \bar{x}_k)^2}}$$

for $i = 1, \ldots, p$ and $k = 1, \ldots, p$.

- $-1 \leq r_{ik} \leq 1$ (it's dimensionless)
- r_{ik} measures the strength of linear relationship
  1. $r_{ik} = 0 \longrightarrow$ no linear association
  2. $r_{ik} < 0 \longrightarrow$ negative linear relationship
  3. $r_{ik} > 0 \longrightarrow$ positive linear relationship

Notations

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1p} \\ \sigma_{12} & \sigma_{22} & \dots & \sigma_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ \sigma_{1p} & \sigma_{2p} & \dots & \sigma_{pp} \end{pmatrix} \quad \mathbf{S} = \begin{pmatrix} s_{11} & s_{12} & \dots & s_{1p} \\ s_{12} & s_{22} & \dots & s_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ s_{1p} & s_{2p} & \dots & s_{pp} \end{pmatrix}$$

$$\mathbf{R} = \begin{pmatrix} r_{11} & r_{12} & \dots & r_{1p} \\ r_{12} & r_{22} & \dots & r_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ r_{1p} & r_{2p} & \dots & r_{pp} \end{pmatrix}$$

Example

Let's use the data from the USDA women's health survey to illustrate some of the descriptive statistics that we just talked about. A random sample of women aged 25-50 years were surveyed for the 5 nutritional components listed below:

- Calcium(mg)
- Iron(mg)
- Protein(g)
- Vitamin A(μg)
- Vitamin C(mg)

We will use the SAS program to carry out the calculations that we would like to see.

Example

Because the SAS output is usually a relatively long document, printing these pages of output out and marking them with notes is highly recommended if not required!

SAS_output_01-eps-converted-to.pdf

Example

The variance-covariance matrix is also copied into the matrix below.

$$\mathbf{S} = \begin{pmatrix} 157829.4 & 940.1 & 6075.8 & 102411.1 & 6701.6 \\ 940.1 & 35.8 & 114.1 & 2383.2 & 137.7 \\ 6075.8 & 114.1 & 934.9 & 7330.1 & 477.2 \\ 102411.1 & 2383.2 & 7330.1 & 2668452.4 & 22063.3 \\ 6701.6 & 137.7 & 477.2 & 22063.3 & 5416.264 \end{pmatrix}$$

- The sample variances are given by the diagonal elements of $\mathbf{S}$.
- The covariances are given by the off-diagonal elements of $\mathbf{S}$.
- Note that, the covariances are all positive, indicating that daily intake of each nutrient increases with increased intake of the remaining nutrients.

Sample Correlations

The sample correlations are included in the table below.

$$\mathbf{R} = \begin{pmatrix} 1.000 & 0.395 & 0.500 & 0.158 & 0.229 \\ 0.395 & 1.000 & 0.623 & 0.244 & 0.313 \\ 0.500 & 0.623 & 1.000 & 0.147 & 0.212 \\ 0.158 & 0.244 & 0.147 & 1.000 & 0.184 \\ 0.229 & 0.313 & 0.212 & 0.184 & 1.000 \end{pmatrix}$$

- Here we can see that the correlation between each of the variables and themselves are all equal to one, and the off-diagonal elements give the correlation between each of the pairs of variables.
- Generally, we look for the strongest correlations first. The results above suggest that protein, iron, and calcium are all positively associated. Each of these three nutrients intake increases with increasing values of the remaining two.

coefficient of determination

The *coefficient of determination* is another measure of association and is simply equal to the square of the correlation. For example, in this case, the coefficient of determination between protein and iron is $(0.623)^2$ or about 0.388.

$$r_{23}^2 = 0.62337^2 = 0.38859.$$

This says that about 39% of the variation in iron intake is explained by protein intake. Or, conversely, 39% of the protein intake is explained by the variation in the iron intake. Both interpretations are equivalent.